

Name: Farooq Abdullah sallam

Subj: Analog Electronic B2IT

Lecturer: Eng: Ala'a Nabil

7 Dec 2020

Analog

Diode

A diode as switch:

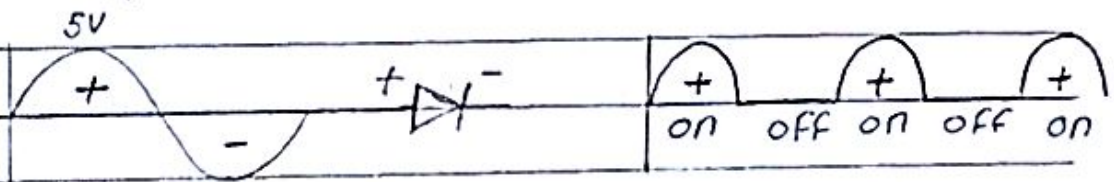
IF D.C is applied $\left\{ \begin{array}{l} \text{always on} \\ \text{always off} \end{array} \right\}$ in this case no switching

note:

IF diode connects with D.C source we cannot control it as switch (on, off)

IF AC is applied:

output



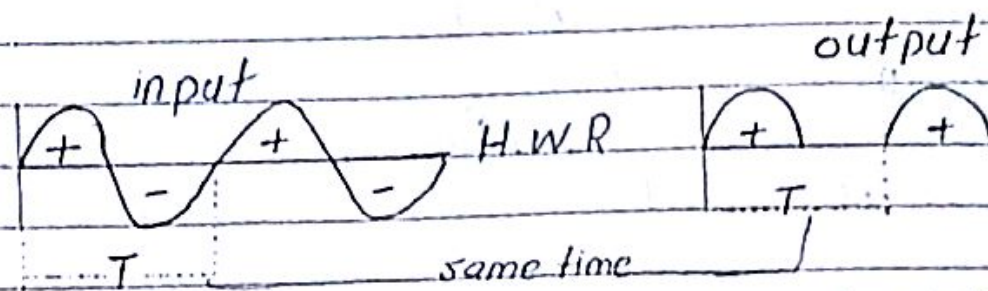
cause of the biasing forward so we got half wave
always the circuit that have one diode the output
will be half wave.

Rectifier:-

A circuit converts AC to DC. it's only be on
when forward bias only.

note:-

The rectifier in its internal design contains of
Diode, resistance, and capacitor... but the most important
part of it is Diode.

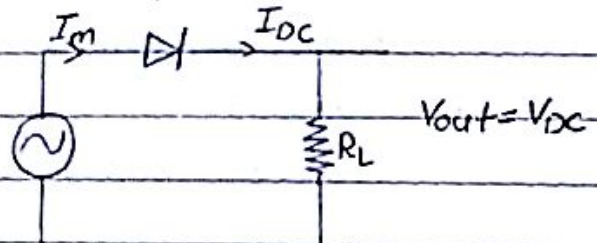


only one diode can be use to get Half wave.

تردد الموجة الخارجة يساوي تماماً تردد الموجة الداخلة ولكن قبل تدخل المكثف. وذلك بسبب أن الزمن متساوي للموجة الداخلة والخارجة المسؤول عن عمل دوره كامله.

Rules that used to find current, voltage and efficiency of ∇ Diode (silicon or Germanium) :-

$$I_m = \frac{V_m - V_D}{R_L + r_f}$$



Where:-

$I_m \rightarrow$ max current

$V_D \rightarrow$ Diode Voltage

$V_m \rightarrow$ max voltage

$r_f \rightarrow$ internal resistance of Diode.

$$I_{DC} = \frac{I_m}{\pi}$$

$I_{DC} \rightarrow$ the output current.

$$V_{DC} = I_{DC} R_L$$

$V_{DC} \rightarrow$ the output voltage after converting operation from AC to DC.

$$\epsilon = \frac{0.406}{1 + \frac{r_f}{R_L}}$$

$\epsilon \rightarrow$ Efficiency.

laws of ideal diode:

$$I_m = \frac{V_m}{R_L} \Rightarrow \text{The internal resistance of ideal diode is zero.}$$

$$I_{DC} = \frac{I_m}{\pi}$$

$$V_{DC} = I_{DC} R_L \Rightarrow V_{DC} = \frac{I_m}{\pi} R_L = \frac{V_m}{R_L \pi} R_L$$

$$C = 0.406 \Rightarrow V_{DC} = \frac{V_m}{\pi}$$

Example:

$$V_{in} = 50 \sin(4\pi t)$$

ideal

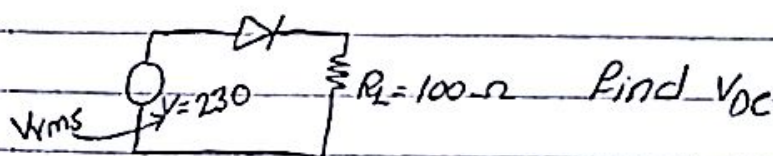
Find output voltage and frequency.

Solu.

$$V_m = 50 \quad \omega = 4\pi$$

$$V_{DC} = \frac{V_m}{\pi} = \frac{50}{\pi} = 15.9 \text{ V}$$

$$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} = \frac{4\pi}{2\pi} = 2 \text{ Hz}$$

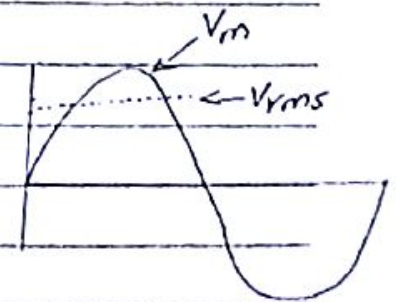


Solu..

$$V_{rms} = 230 \text{ V}$$

$$V_m = V_{rms} \times \sqrt{2} = 230 \times \sqrt{2} = 325.3 \text{ V}$$

$$V_{DC} = \frac{V_m}{\pi} = \frac{325.3}{\pi} = 103.6 \text{ V}$$



$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

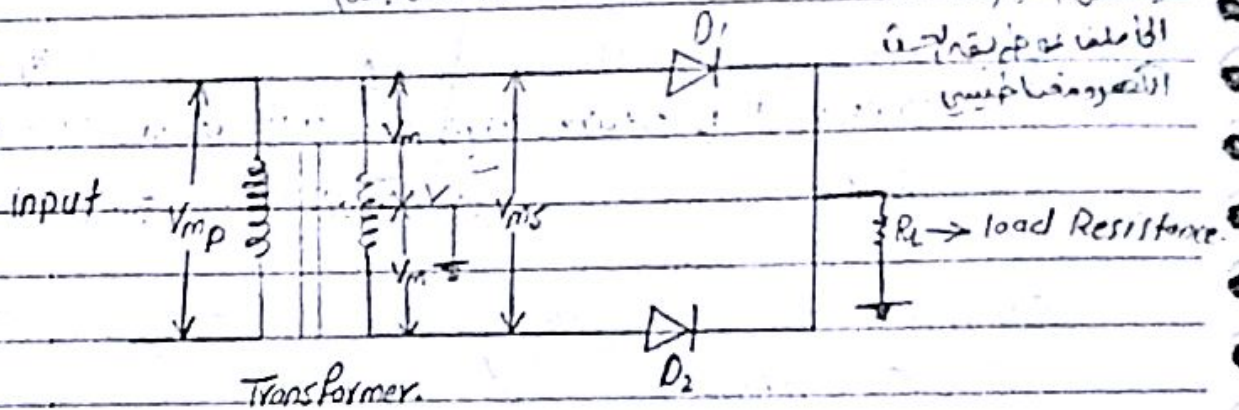
Full wave rectifier

Def of FWR:

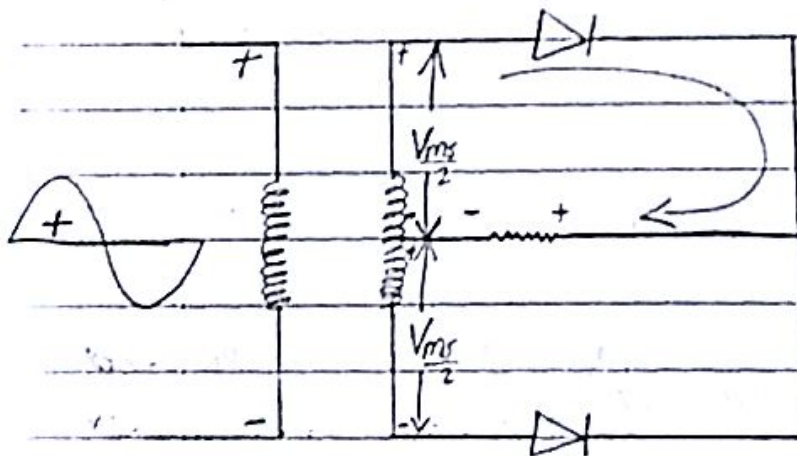
is the current that's flowing for both half cycle of the input.

1) Center tap

يقسم جهد الجان
الى اثنى عشر باقسام

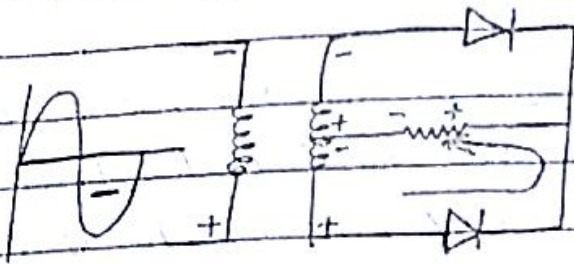


For positive half cycle of input:



هذه الة ايود بها انه لن يشغل
يجب عليه ان يكون قادر على حمل اثنى عشر باقسام
تبع $V_m/2$ كامله.

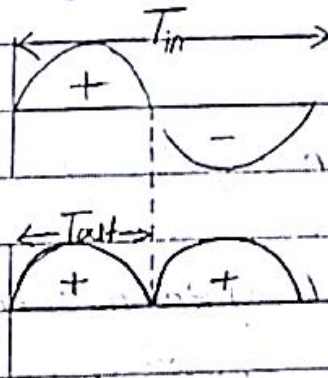
for negative half cycle input:



ففي هذه الحالة الدايود السفلي يفتح
الذي يمتثل، ولعلني يجب عليه لتحل
بفولتية، انعكس كالمه.

والرخصة:

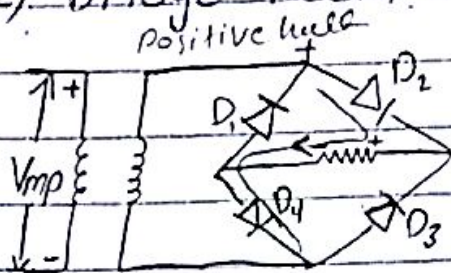
الفرق بين الخارجة في الحالتين positive half & negative half
تكونان نفس القطبية وهذا ما نلاحظه من الرسمين السابقين



$$\therefore T_{out} = T_{in}/2$$

$$\therefore F_{out} = 2F_{in}$$

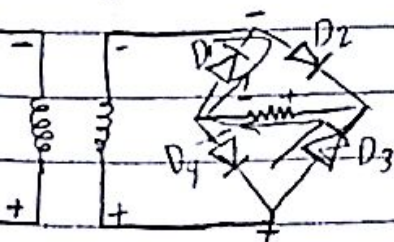
2) Bridge Rectifier:



For positive:

في هذه الحالة الدايود الذي يمتثل
هم D_1 و D_3 وسكون حركة التيار
كما هو موضح في الرسم

negative half



ايضا في ال Bridge Rectifier قطبية
ال output تكون متساوية

Follow.

The most differences between the center tap and Bridge rectifier.

The PIV of Bridge rectifier is less than the PIV of center because in the Bridge rectifier the V_{ms} distributed between 2 diodes while in center the PIV for one diode only.

The V_{out} of Bridge rectifier is larger than the V_{out} of center because in bridge there is no center tap divide while in center rectifier there is center tap divide the V_{ms} between 2 diodes. $\rightarrow$ the V_{ms} .

e.g)

in the center tap circuit show in fig. the diodes are assumed to be ideal, having zero internal resistance.

Find) 1) d.c output v. 2) PIV 3) Rectifier efficiency.

Solu.

$$V_{rms} = 230V$$

$$V_{mp} = 230 \times \sqrt{2} = 325.3V$$

$$V_{ms} = V_{mp} \frac{N_2}{N_1} \Rightarrow V_{ms} = 325.3 \times \frac{1}{5} = 65V$$

$$V_{ms} = \frac{V_m}{\sqrt{2}}$$

$$\text{For center tap} = \frac{65}{2} = 32.5V$$

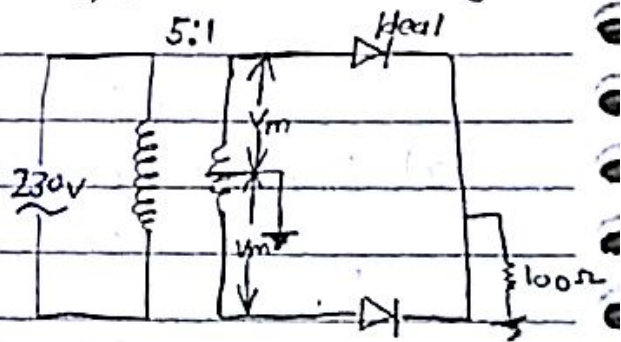
$$\therefore V_m = 32.5V$$

$$(1) V_{DC} = \frac{2V_m}{\pi} = \frac{65}{\pi} = 20.7V$$

$$(2) PIV = 65V = V_{ms}$$

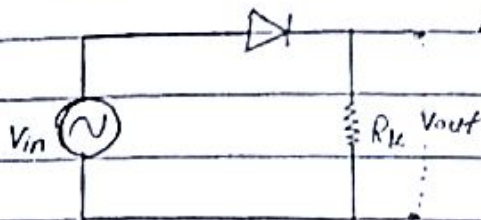
$$3) \eta = 0.812 = 81.2\%$$

5:1 transformer ratio
2 diodes in series



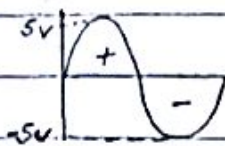
clipper

is a circuit that clips off a portion of the input wave form.

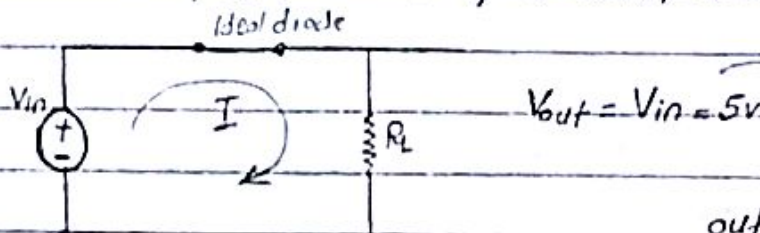


Noting:-

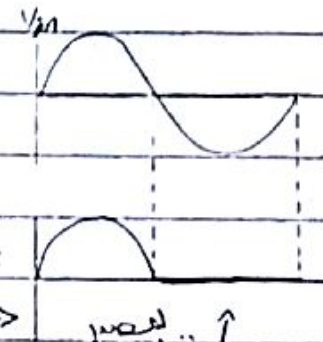
When the diode is series with the source and it's on so then there will be an output.



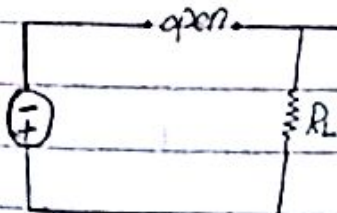
For positive half cycle of input:-



output wave



For negative half cycle of input:-

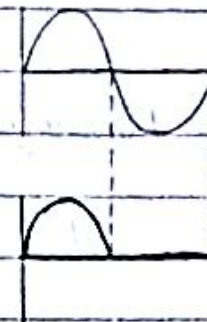


in this case

$V_{out} = 0$ cause there's no current flowing in circuit.

التيار لا يمر
لأنه في هذه الحالة
الديود يكون كدائرة مفتوحة
open circuit
لذلك يكون الناتج صفر

output

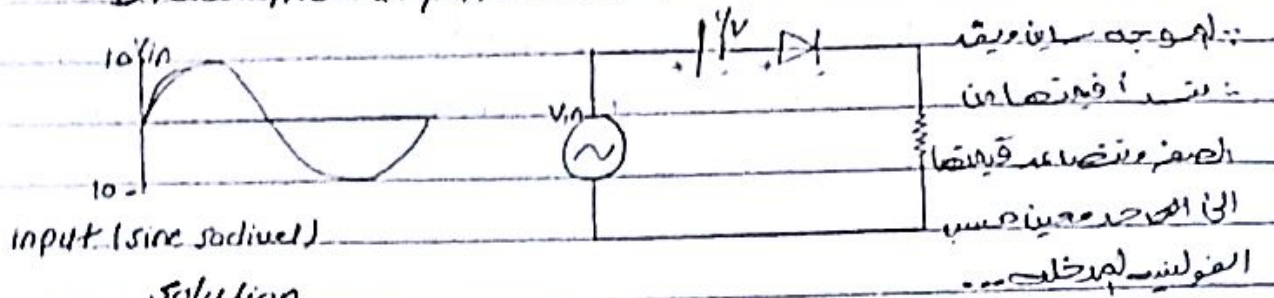


لأنه الذاة صغرى مقصودة ليكون التيار لا يمر
في هذه الحالة اذا كان متصل على البطارية

Follow.

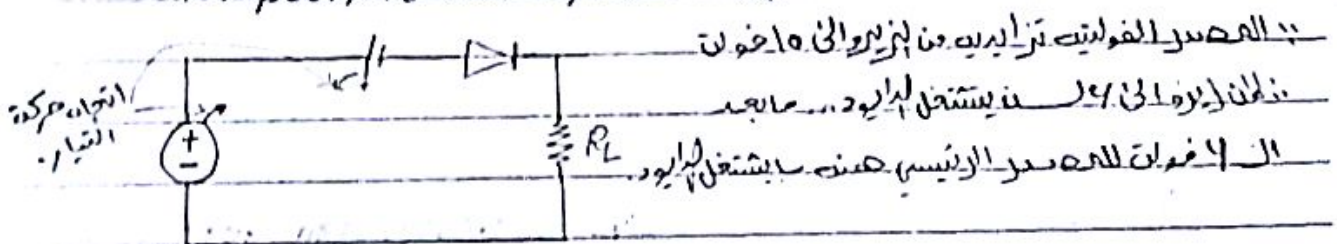
في هذا المثال... نلاحظ ان الدايود موصول بين DC و AC. في هذه الحالة
مع اشارة AC عكسي مع DC. لذلك في هذه الحالة AC يتبادل اشارة
بشغلة وال DC يتبادل اشارة في هذه الحالة... نشوف اولاً عن الدايود ما دونه اشارة

Draw the output waveform:-



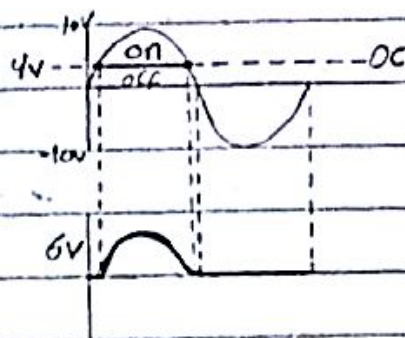
Solution..

For positive half cycle of input



When $V_{in} < 4V$: Diode off $\rightarrow V_o = 0$

When $V_{in} > 4V$: Diode on $\rightarrow V_o = V_{in} - 4 = 10 - 4 = 6V$

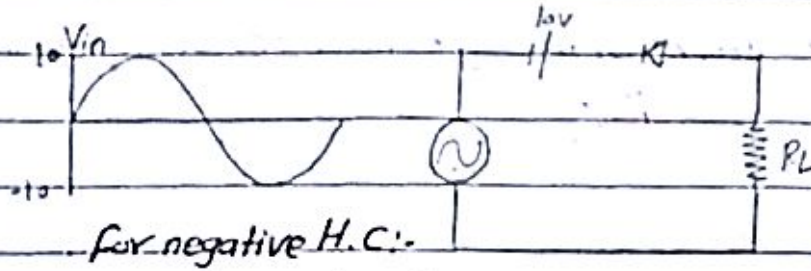


For negative half cycle:-

Diode is off $\rightarrow V_o = 0$

لأنه يتكون الدايود من عازل... ربط عكسي مع المصدر... لذلك الفولتية الخارجة تكون (0)

Follow.

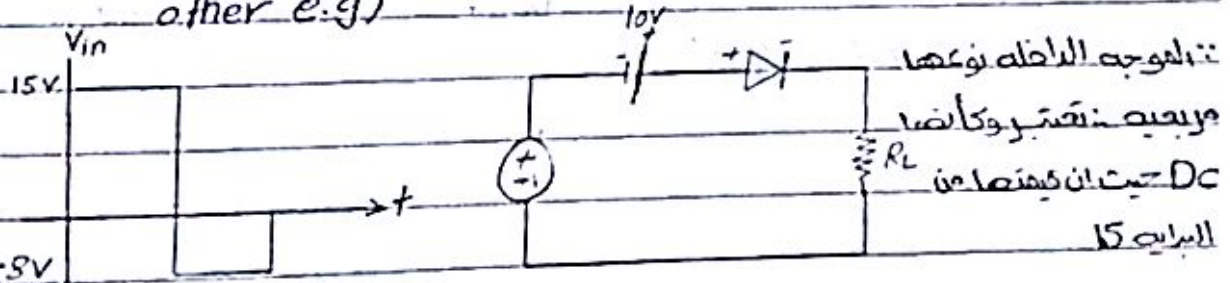


While the diode voltage of the AC and DC are equal so the output voltage will be 0. $V_o = V_{ac} - V_{dc} = 10 - 10 = 0$

for positive H.C:-

While the diode is in reverse bias with the two sources so the V_{out} will be 0 because the circuit will be as an open circuit.

other e.g)



الموجة الاشارة
موجبة منقطه وكذا
DC حيث ان قيمتها
الباريه 15

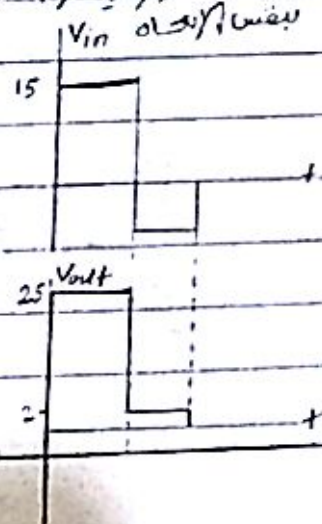
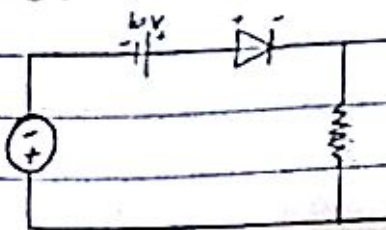
Solu..

For positive half cycle:-

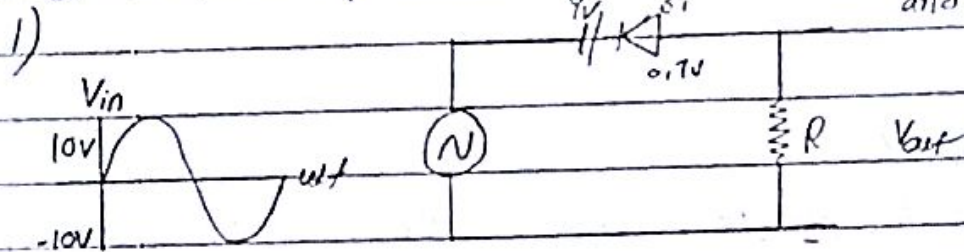
Diode is on $\rightarrow V_{out} = V_{ac} + V_{dc} = 15 + 10 = 25V$

For Negative half cycle:-

Diode is on $\rightarrow V_{out} = V_{dc} - V_{ac} = 10 - 8 = 2V$

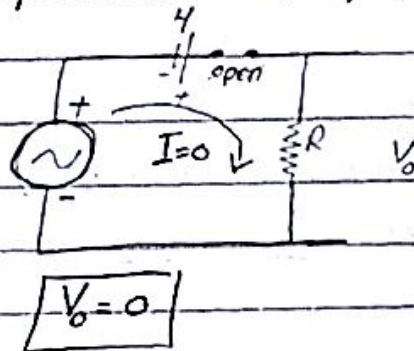


Follow.

Draw the output waveform. The diode series with source and reverse with D_c .

Soln.

For positive half cycle



للموجبة الموجبة تكون

التيار في الدارة مفتوحة

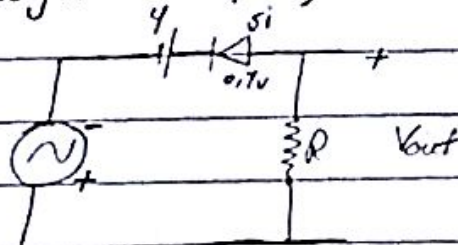
في هذه الحالة يكون

التيار في الدارة يساوي صفر

في هذه الحالة يكون

التيار في الدارة يساوي صفر

For negative half cycle:



هنا يكون التيار في الدارة

يساوي صفر

في هذه الحالة يكون

When $V_{in} < 4.7$ the diode is off $V_o = 0$ When $V_{in} \geq 4.7$ the diode is on

Now we'll calculate the magnitude of output voltage:

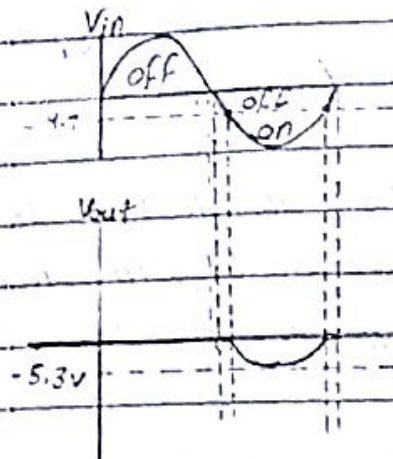
by using K.V.L:-

$$V_{out} - 0.7 - 4 + 10 = 0$$

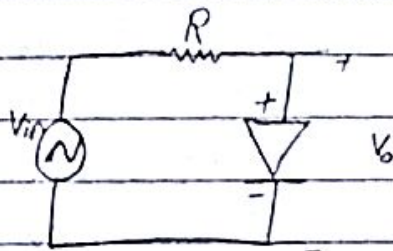
$$V_o = -4.7 + 10$$

$$V_o = -5.3V$$

Follow.



2) Diode connected parallel with source:-



If the Diode is ideal and connected parallel with source then the V_{out} directly equal zero. cause it's being like short circuit

Note:-

If the Diode is ideal and connected parallel with source then:- When ideal diode is Forward:-

short circuit

$$V=0$$

$$R=0$$

I_V that's mean there's a current in that wire.

When ideal diode is Reverse:-

open circuit

$I=0$ cause it's open circuit

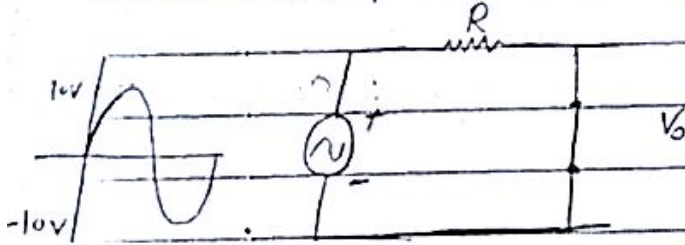
$V_o = V_{in}$ that's mean there's a voltage and it will be the equal to V_{in}

$$V_o = V_{in}$$

Follow.

$$V = I \cdot R \quad I = 0 \Rightarrow 0$$

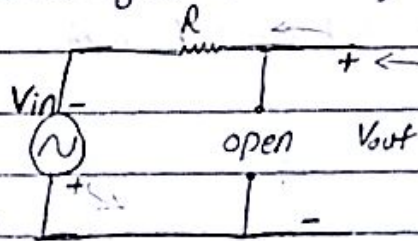
for positive half cycle:



While the diode is ideal, forward and parallel with source so

$$V_{out} = 0$$

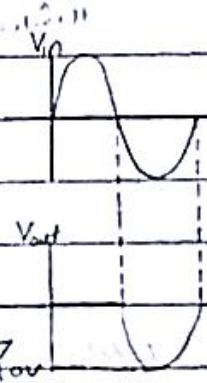
for negative half cycle:-



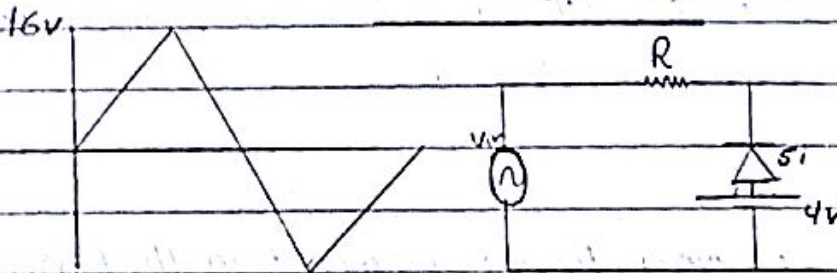
إذا كان
بعد
مفصل
بالقائه
إذا كان
نفسه بالقائه

$$V_{out} + 10 = 0$$

$$V_{out} = -10V$$

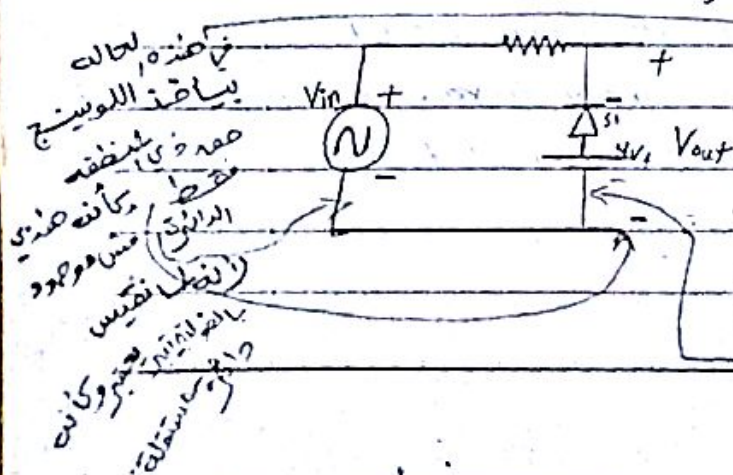


هذه
سواء
التي



Draw the wave form of V_{out} :-

for positive half cycle:-



When $V_{in} < 3.3$ Diode is on

$$V_{out} + 0.7 - 4 = 0$$

$$V_{out} = 4 - 0.7 = 3.3V$$

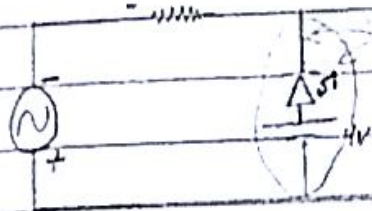
لا يجب
عشان
البطارية
فقط

When $V_{in} > 3.3$ Diode is off

$$V_{out} - V_{in} = 0 \Rightarrow V_{out} = V_{in}$$

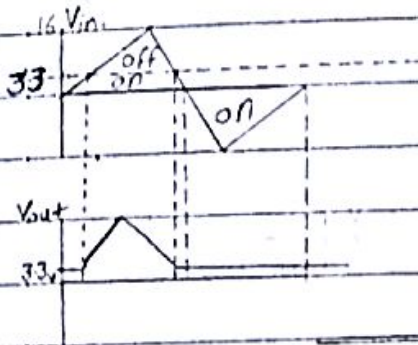
Follow.

for negative half cycle:-

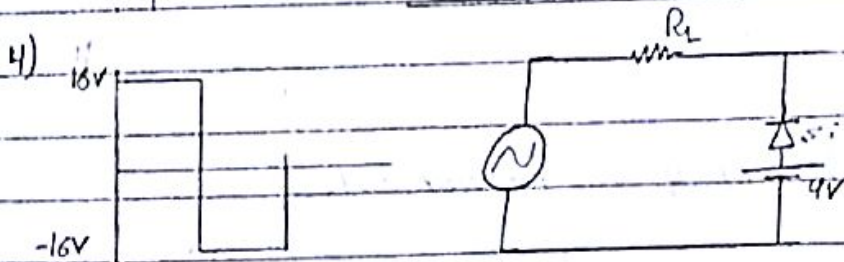


Diode is on

$$V_{out} = 3.3V$$



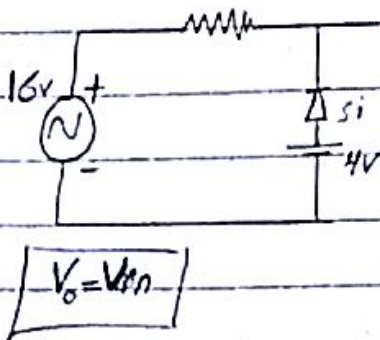
← The output wave.



same solu...

for positive half cycle:-

Diode is off.

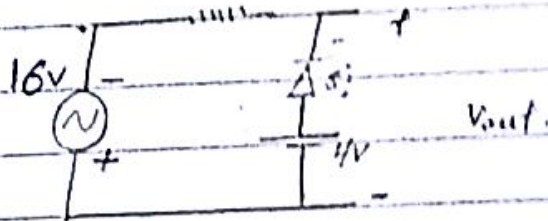


Directly the $V_o = V_{in}$ cause the controller from the beginning is the AC square form. cause it starts with 16V and it's connected reverse with Diode.

$$V_o = V_{in}$$

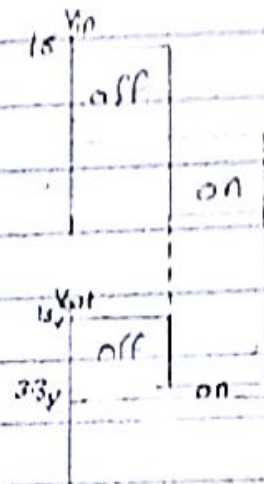
Reflex

For negative half cycle:

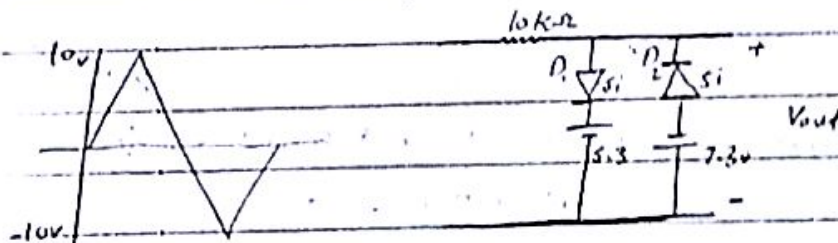


$$V_{out} + 0.7 - 4 = 0$$

$$V_{out} = 3.3V$$



Sketch the I_R and V_o for the network:



المخرج = 1.8V
التيار المار في المقاومة = 0.6mA

Solution:

For positive half cycle:

in this case the D_1 will be off cause it will be in reverse bias with the AC and DC

When $V_{in} < 6$ D_1 is off and $V_o = V_{in}$.

When $V_{in} \geq 6$ D_1 is on and $V_{out} = 0.7 + 5.3 = 6V$

$$I_R = \frac{V_{out} - 6}{R_L} = \frac{6 - 6}{10^4} = 0.6mA$$

Follow.

for negative half cycle:-

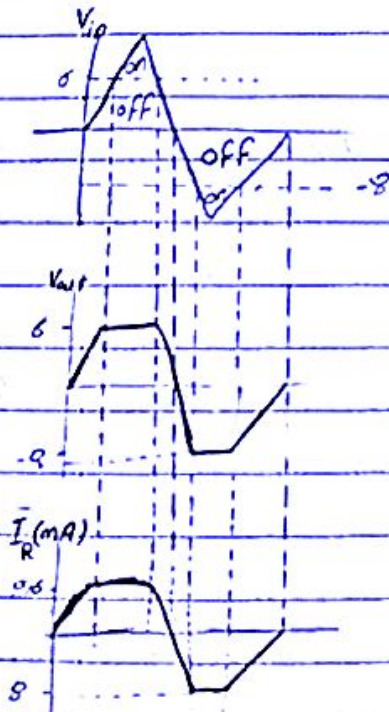
D_1 Will be off cause it will be in reverse with AC and DC

فولتية البطارية مع فولتية الـ AC

When $V_{in} < 8$, D_2 is off and $V_o = V_{in}$

When $V_{in} \geq 8$, D_2 is off and $V_o = -0.7 - 7.3 = -8V$

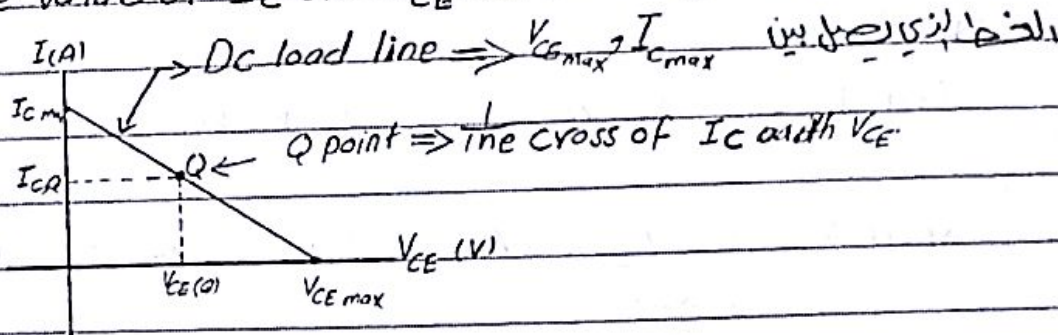
$$I_R = \frac{V_{out}}{R_L} = \frac{-8}{104} = -0.8 \text{ mA}$$



ملاحظة ١-
 وحدة الـ ٨ في "حرفياً" تكون
 مشابهة للفولتية ٨ في كل
 لحظة... يكون التيار "٨ في كل لحظة" امين
 فكل لحظة فكل لحظة "٨" "٨"

operating point of DC load line

operating point: DC load line

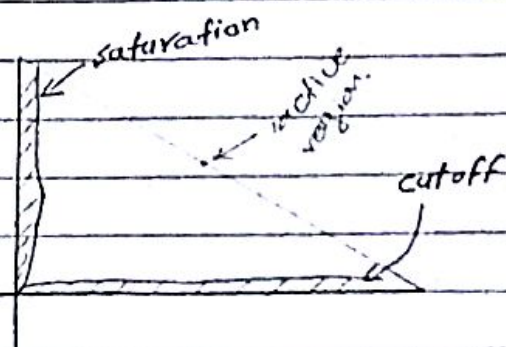
The value of I_C and V_{CE} when no signal is applied to a transis.

location of point:-

- 1) Cutoff region:- When $I_C \approx 0$ "Transistor acts off switch"
- 2) Saturation region:- When I_C reaches maximum value, "transistor acts on switch"
- 3) Active region:-
The region between cutoff and saturation. "Transistor acts as amplifier."

Note:-

If we want to get the value of $V_{CE\max}$ all we must do just put $I_C \Rightarrow \text{zero}(0)$, and if we want to get $I_{C\max}$ we must put $V_{CE} \Rightarrow \text{zero}(0)$

 $V_{CE\max} \Rightarrow \text{put } I_C = 0$ $I_{C\max} \Rightarrow \text{put } V_{CE} = 0$ 

Transistor biasing:-

Dc voltage applied to a transistor to control the two transistor junctions.

Diode biasing:-

Dc voltage applied to a diode to control diode junction.

The purpose of transistor biasing:-

- To ensure that, the transistor operate in the desired region.

كشأن ضمن انه سيكون شغال
بالمنطقة المرغوب فيها سواء
Cutoff
Saturation
Active

Transistor stability:-

The process of making Q point independent of temperature change and variation in transistor parameter (β)

Q point can be shifted to undesired region by the following factors:-

a) Change in temperature.

b) Variation in parameter (β) $\Rightarrow \beta = \frac{I_c}{I_B}$

Thermal runaway:-

The further increasing in temperature leading to transistor destruction.
heat sink $\leftarrow$ 

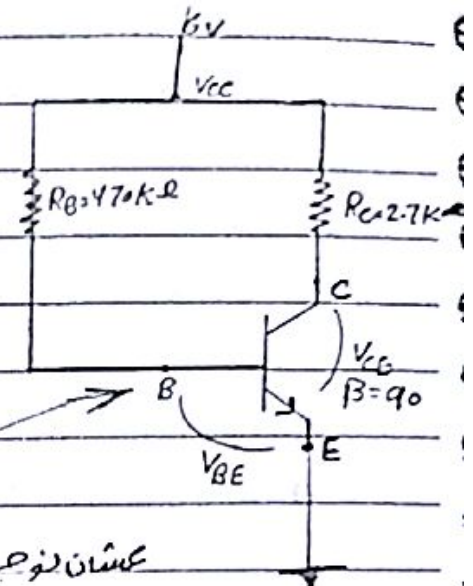
Follow.

Transistor Biasing methods

1) Base transistor method (Fixed bias)

Find from the following circuit:-

$$I_B, I_C, V_{CE} \text{ and } V_B \quad B=90 \text{ Given}$$



to find I_B and V_B we do that:-

$$V_{CC} - I_B R_B - V_{BE} = 0 \quad \leftarrow \text{First loop}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 - 0.7}{470K} = 0.03255 \text{ mA}$$

$$V_B - V_{BE} = 0 \Rightarrow V_B = V_{BE} = 0.7 \quad \leftarrow V_B \text{ كمان نو صر ال نو صر م قضيت}$$

$$V_{CC} - I_B R_B - V_B = 0$$

$$V_B = V_{CC} - I_B R_B$$

$$V_B = 16 - (0.03255 \text{ mA} \times 470K)$$

$$V_B = 0.7$$

الاولى انه نبدأ من عند V_{CC} ونطرح ايفولوشيان الذي نواجهه الى ان نصل الى V_{BE} فطوله ولوقت الطريقة الثانية وهي اسهل انه نبدأ من عند V_B نفسه ونطرح ما بعده وانا نتج يكون نفسه.

second loop:-

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\beta = \frac{I_C}{I_B}$$

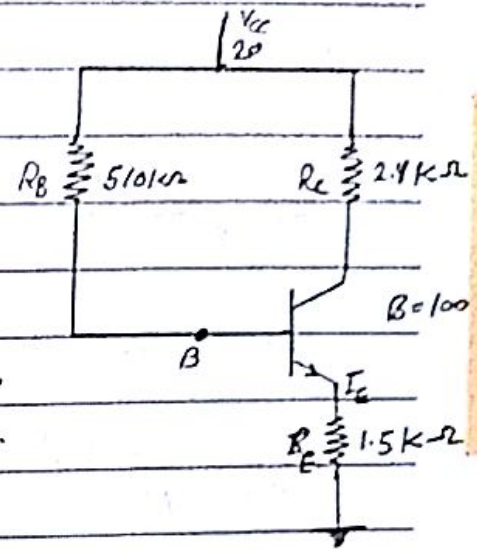
$$I_C = \beta I_B = 90 \times 0.03255 = 2.92 \text{ mA} \quad \leftarrow \text{في المعادلة اعلاه نو صر مخصصين صما (} V_{CE}, I_C \text{) نو صر } I_C \text{ ال } \beta \text{ معتمداً على } I_C$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = 16 - (2.92 \text{ mA} \times 2.7K) = 8.1 \text{ V}$$

Follow.

Deter the Following:- and Draw the
 $I_B, I_C, V_{CE}, V_E, V_C$ Dc load line.
 and Q point.



First loop:-

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

لحل الدارة
 نستخدم قانون كيرشوف للجهد

$$I_E = (\beta + 1) I_B \approx \beta I_B$$

نستخدم العلاقة بين
 التيارين I_E و I_B

$$I_E \approx I_C$$

$$V_{CC} - I_B R_B - V_{BE} - \beta I_B R_E = 0$$

$$I_B (R_B + \beta R_E) = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_E} = \frac{20 - 0.7}{510 + (100 \times 1.5)} \text{ k} = 0.0292 \text{ mA}$$

second loop:-

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0 \Rightarrow \boxed{I_E \approx I_C}$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0 \Rightarrow I_C = \beta I_B \Rightarrow I_C = 100 \times 0.0292 =$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) \quad I_C = 2.92 \text{ mA}$$

$$V_{CE} = 20 - 2.92 \text{ mA} (2.4 + 1.5) \text{ k} = 8.6 \text{ V}$$

Now we'll find V_E :-

$$V_E - I_E R_E = 0 \quad V_E = 2.92 \text{ mA} \times 1.5 \text{ k} = 4.4 \text{ V}$$

$$\boxed{I_E \approx I_C}$$

Now we'll find V_C :-

$$V_{CC} - I_C R_C - V_C = 0$$

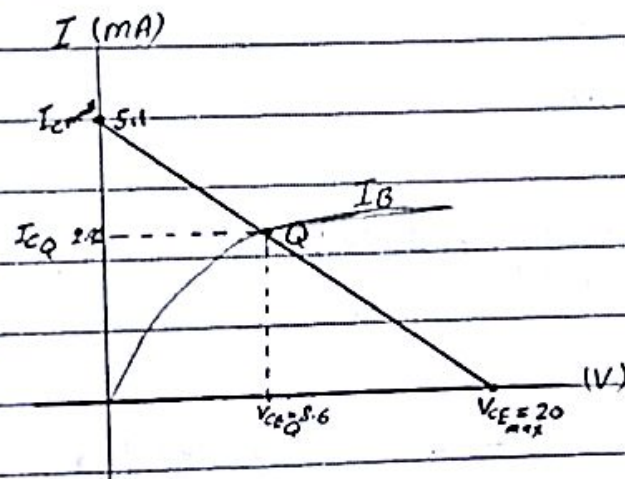
$$V_C = 20 - (2.92 \text{ mA} \times 2.4 \text{ k}) = 13 \text{ V}$$

to draw the DC load line we must find $I_{C\text{-max}}$, $V_{CE\text{-max}}$

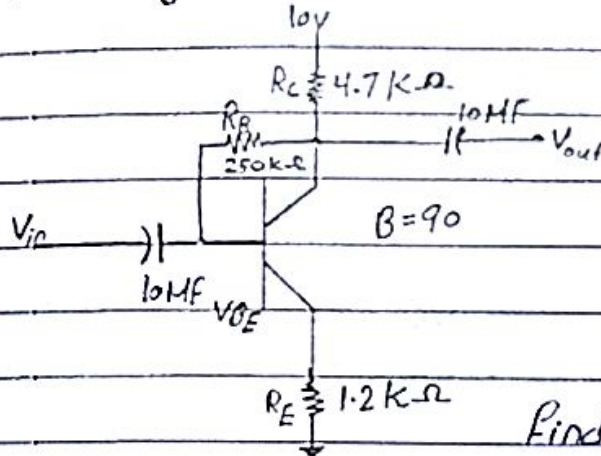
$$V_{CE\text{-max}} = V_{CC} - I_C(R_C + R_E) \quad \text{put } \boxed{I_C = 0} \text{ to find } V_{CE\text{-max}}$$

$$V_{CE\text{-max}} = V_{CC} = 20 \text{ V}$$

$$I_{C\text{-max}} = \frac{V_{CC} - V_{CE}}{R_C + R_E} \quad \text{put } \boxed{V_{CE} = 0} \text{ to find } I_{C\text{-max}}$$
$$I_{C\text{-max}} = \frac{20}{(2.4 + 1.5) \text{ k}} = 5.1 \text{ mA}$$



Biasing with feedback resistor:-



Find I_C and V_{CE}

First loop:-

I_B and I_C relation

$$V_{CC} - I_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - \beta I_B R_C - I_B R_B - V_{BE} - \beta I_B R_E = 0$$

~~$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$~~

$$\beta = \frac{I_C}{I_B} \Rightarrow I_C = \beta I_B$$

$$\therefore I_E \approx I_C$$

$$\therefore I_E = \beta I_B$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{10 - 0.7}{250k + 90(4.7k + 1.2k)} = 0.0119 \text{ mA}$$

$$I_C = \beta I_B = 90 \times 0.0119 = 1.071 \text{ mA}$$

Second loop:-

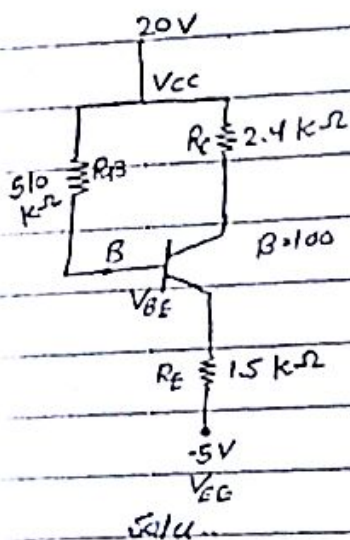
$$V_{CC} - R_C I_C - V_{CE} - R_E I_E = 0 \quad \rightarrow R_E \approx R_C$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 10 - 1.071 \text{ mA} (4.7k + 1.2k)$$

$$V_{CE} = 3.68 \text{ V}$$



Find I_C and V_{CE} and then draw the load line.

$$I_B = \frac{V_{CC} - V_{BE} - V_{EE}}{R_B + R_E \beta} = \frac{20 - 0.7 - 5}{(510 + 15 \times 100) \text{ k}\Omega} = 0.0362 \text{ mA}$$

$$I_C = \beta I_B = 100 \times 0.0362 = 3.62 \text{ mA}$$

$$V_{CC} - I_C(R_C + R_E) - V_{CE} - V_{EE} = 0$$

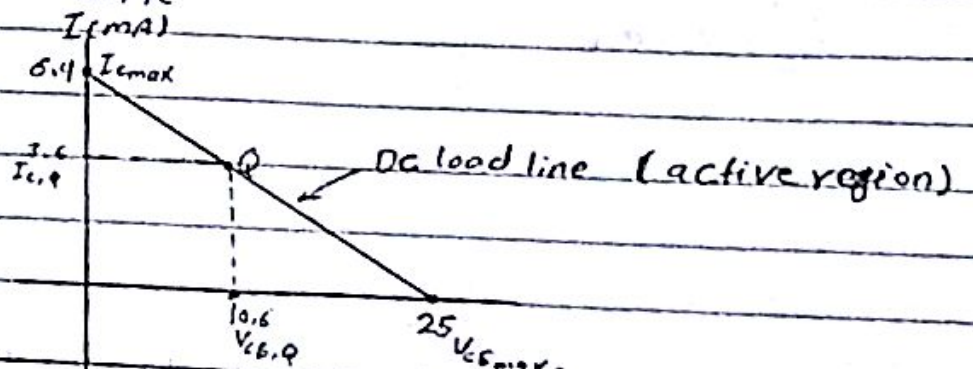
$$V_{CE} = V_{CC} - I_C(R_C + R_E) - V_{EE} = 20 - 3.62 \text{ mA} \times (2.4 + 1.5) \text{ k}\Omega - 5$$

$$\therefore V_{CE} = 10.6 \text{ V}$$

To draw DC load line we must get $I_{C \text{ max}}$ & $V_{CE \text{ max}}$.

$$V_{CE \text{ max}} = V_{CC} - V_{EE} = 25 \text{ V}$$

$$I_{C \text{ max}} = \frac{V_{CC} - V_{EE}}{R_C + R_E} = \frac{25}{3.9 \text{ k}\Omega} = 6.4 \text{ mA}$$



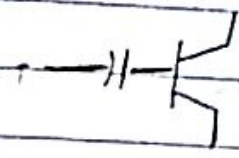
Capacitor

Transistor types:-

1) $C_{in} \rightarrow$ coupling capacitor.

Coupling capacitor:-

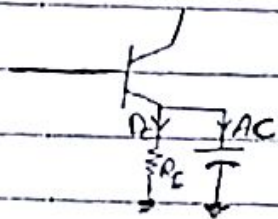
it's used to decouple AC and DC so as not disturb Q point when AC is applied.



2) bypass:-

استخدم لتوفير مسار مقاومة منخفضة لـ AC

it's used to provide a low impedance path for AC.



نلاحظ ان الكثافة على القاربي مع المقاومة

وذلك لتتوزع فيه التيار بين الـ AC , DC

حيث ان AC يذهب باتجاه الكثافة لانه

بالنسبة له المقاومة السعوية للكثافة تترى صفر

$$X_C = \frac{1}{2\pi f} \approx 0 \quad \text{very high}$$

لذلك يختار الطريقة التي ما يفقد من قيمته

فيما التيار المستمر الذي يمر بالكثافة وذلك لانه بالنسبة له الكثافة open circuit

لانه يمكن مقاومة الكثافة (المقاومة السعوية) ما لا نهائية وليس فان يكون المقاومة

ما لا نهائية وهذا يعني ان الجهد موصول توصيل عكسي لانه ان يمر التيار المستمر

$$X_C = \frac{1}{2\pi f} = \frac{1}{0} = \infty$$

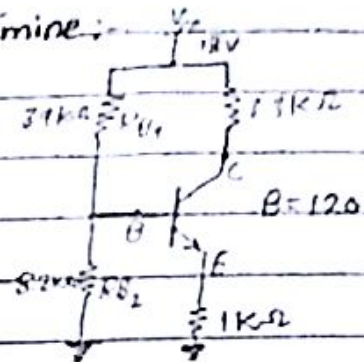
تزداد التيار المستمر يساوي صفر وهذا يعني معروفة

For the voltage divider bias, determine:

I_C and V_{CE}

Solu.

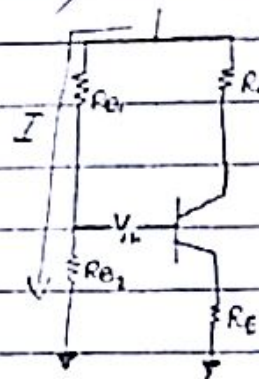
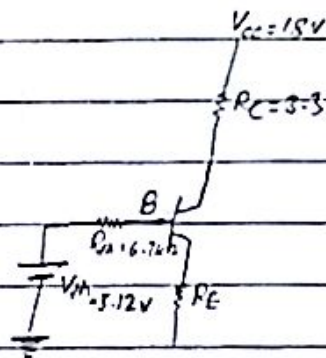
$$I = \frac{V_{CC}}{R_{B1} + R_{B2}} = \frac{18}{47.2K} = 0.38 \text{ mA}$$



$$V_{Th} = I * R_{B2} = \frac{V_{CC} * R_{B2}}{R_{B1} + R_{B2}} = 0.38 \text{ mA} * 8.2K = 3.116 \text{ V}$$

$$R_{Th} = \frac{R_{B1} * R_{B2}}{R_{B1} + R_{B2}} = 6.7K \Omega$$

equivalent Thevenins circuit:



$$V_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

من این یه ایشنل عشان
نه جدره طالب.

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + \beta R_E} = \frac{3.12 - 0.7}{(6.7 + 120 * 1)K} = 0.0191 \text{ mA}$$

$$I_C = \beta I_B = 120 * 0.0191 = 2.29 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) = 18 - 2.29 \text{ mA} * (3.3 + 1)K = 8.15 \text{ V}$$

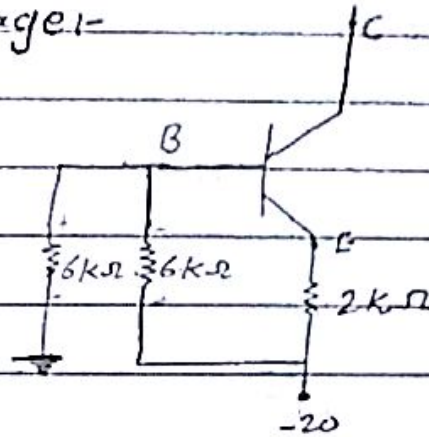
Calculate the base voltage:-

Solu.

$$I = \frac{V_{CC}}{R_B + R_E}$$

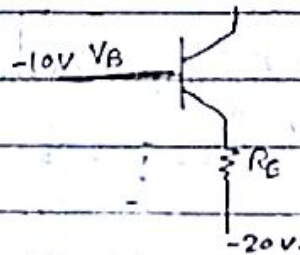
$$V_B - V_{BE} - I R_E - V_{EE} = 0$$

$$V_B = \frac{-20 + 6}{6 + 6} = -10V$$



$$V_B - V_{BE} - I R_E - V_{EE} = 0$$

$$I = \frac{-10 - 0.7 + 20}{2K} = 4.65mA$$



7

Find the I_C , V_{CE}

solu.

$$I = \frac{V_{CC}}{R_{B1} + R_{B2}} = 0.225 \text{ mA}$$

$$V_{Th} = I * R_{B2} = 0.225 \text{ mA} * 91 \text{ K} = 20.48 \text{ V}$$

$$R_{Th} = \frac{R_{B1} * R_{B2}}{R_{B1} + R_{B2}} = 7.93 \text{ K}\Omega$$

to find I_C , first I must find I_B

$$V_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

$\uparrow$
 $B I_B$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + \beta R_E} = \frac{20.48 - 0.7}{(7.93 + (80 * 0.68)) \text{ K}} = 0.0216 \text{ mA}$$

$$I_C = \beta I_B = 80 * 0.0216 = 1.728 \text{ mA}$$

Now I'll find V_{CE}

$$V_{CE} = V_{CC} - I_C (R_C + R_E) = 16 - 1.728 (3.9 + 0.68) \text{ K} = 8.08 \text{ V}$$

